2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

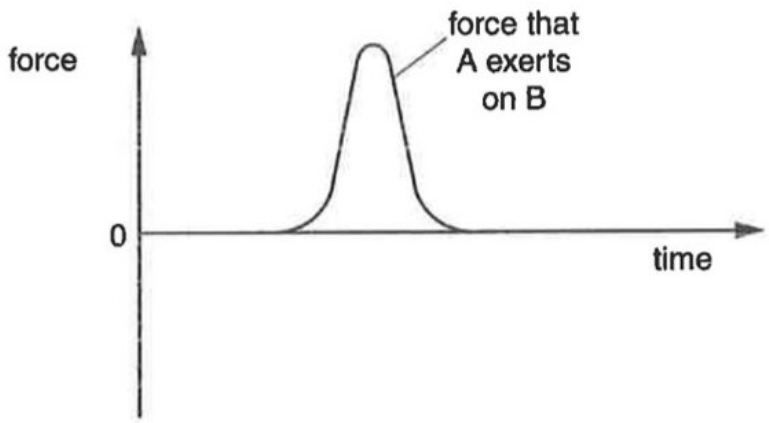


Fig. 2.1

(a) **(i)** Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

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..... [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.

Calculate the change in velocity of

(i) the car,

change in velocity = ... $m\ s^{-1}$ [2]

(ii) the truck,

change in velocity = ... $m\ s^{-1}$ [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

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..... [2]

[Total: 8]